

# Jannatul Tuna Trisha

Computer Science and Engineering Student

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## Profile

Computer Science and Engineering student at Southeast University with hands-on experience in web development, software projects, machine learning, healthcare AI, embedded systems, and robotics. Interested in building practical, well-documented applications and contributing to academic projects, research opportunities, and technology-based solutions.

## Education

### B.Sc. in Computer Science and Engineering

Southeast University, Dhaka, Bangladesh

2023 – Present

Current semester: 9th

### Higher Secondary School Certificate (Science)

Barisal Government Women's College

2021

### Secondary School Certificate (Science)

Barishal Government Girls High School

2019

## Technical Skills

**Programming** C, C++, Java, Python, JavaScript

**Web Development** HTML5, CSS3, React, Node.js, Express.js, PHP, MySQL, responsive design

**Machine Learning** Scikit-learn, Pandas, NumPy, XGBoost, LightGBM, ensemble learning, SHAP, LIME

**Tools & Platforms** Git, GitHub, VS Code, Jupyter Notebook, Linux, XAMPP, Arduino IDE

## Selected Projects

### MindMate

Full-stack digital wellness application featuring authentication, journaling, mood tracking, AI-powered chat, and activity statistics.

*Technologies: React, Node.js, Express, MongoDB.*

[GitHub](#)

### Car Rental System

Java console application for vehicle availability, customer management, rentals, returns, and rental-cost calculation, developed using object-oriented programming.

[GitHub](#)

### SEU Tech Event Registration Page

Responsive website presenting technology-event information, a schedule, and a student registration form using semantic HTML and CSS.

[Coursework](#)

### Competition Fast Line Follower Robot

Autonomous robotics project for stable, high-speed line tracking through sensor-based detection, motor correction, and PID control. *Technologies: Embedded C/C++, IR sensors, Arduino IDE.*

Ongoing

## Research-Oriented Work

- **Leukemia Screening Using CBC Data:** Leakage-controlled machine-learning study focused on patient-aware evaluation, site variation, robustness, and explainable AI.
- **Obesity Risk Classification:** Multi-class classification using baseline models, hyperparameter optimization, soft-voting, and stacking ensembles.
- **Life Expectancy Prediction:** Regression study comparing model generalization through random testing, country-disjoint validation, and future-year evaluation.
- **Cancer Risk Prediction:** Exploration of lifestyle and health-related factors using preprocessing, model comparison, ensemble learning, and explainability.

## Additional Strengths

Problem solving, software testing, technical documentation, REST APIs, model validation, and explainable AI.